

Where should the AI run?

Sir asked: can we apply this research on the flight controller, or through Mission Planner?
Which is easier?

Neither. The easy and correct place is a companion computer with PX4.

The flight controller only flies the drone. Mission Planner is ground software for a different autopilot. Our AI needs a small Linux board on the drone.

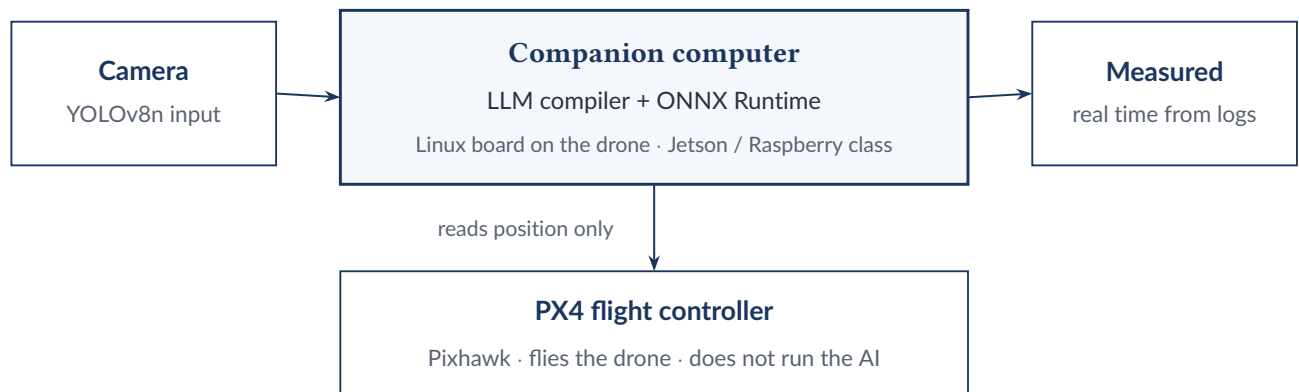
A **flight controller** (Pixhawk) is the tiny chip that keeps the drone in the air. A **companion computer** is an extra Linux board for heavier work such as cameras and neural nets. **Mission Planner** is laptop software, mainly for ArduPilot. This project already uses **PX4**.



Three places people might put the AI. Only the companion computer on the drone is both possible and on the UAV.

How the work is split

Atul's notes put it simply: the flight controller is not strong enough for video or AI, so the drone needs a second computer. The LLM only thinks about compile choices. It does not fly the drone.



Same split as the notebook: camera and neural net on the companion; Pixhawk only flies. Groq may suggest compile strategies. It is not the measured run. If a ground screen is needed later, use QGroundControl to watch PX4 — do not put the compiler there.

Today, and later



Already working in simulation: ONNX compiler, YOLOv8n, ROS inference on the Linux host, PX4 SITL with live position. TensorRT is skipped here because this machine has no NVIDIA GPU.

For the thesis

We continue with companion + PX4. We will not put the compiler inside flight-controller firmware. We will not build a Mission Planner plugin.

The LLM may only suggest allowed compile strategies. The compiler checks and compiles. The profiler measures real time from logs. Telemetry only reads position. The AI does not take over flying.